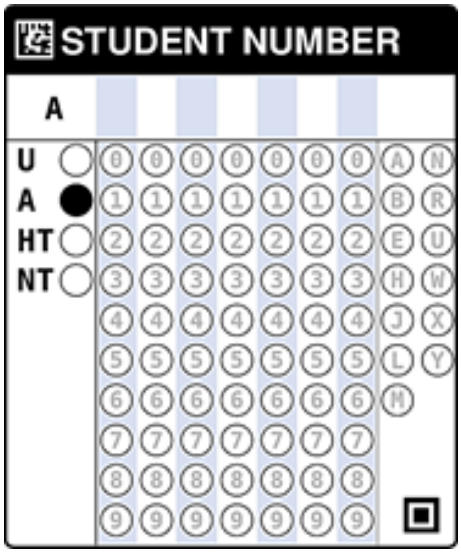
The Answer Sheet for Semester 1 AY2025/26

Write your Student Number in the box below using **(2B) pencil**. **Do NOT write your name**.



Section	Maximum Marks	Your Marks	Grading Remarks
Total	100		

Box A.1. Grid Problem

Special case 1: return grid[0][0] * k
Special case 2: return sum(k largest elements)

Idea:
Use greedy algorithm. For each row i, we find the limits[i] largest elements and add them to a candidate list l. Then we take the k largest elements from l and return the sum.

Justification:
For each row i, we can only take at most limits[i] elements, and to maximise the sum we should always take the largest elements available. And in the optimal solution, all k selected elements must come from these elements.

Time complexity:
For convenience, denote l_i = limits[i] and s = sum(limits).

(i) Naive approach: We can first sort the m rows each of size n, which is O(mn log n). Then we sort the candidate list l and take the k largest elements, which is O(s log s). Thus the overall time complexity is O(mn log n + s log s). Use merge sort to get worst case guarantees.

(ii) A slightly better approach: We can maintain min-heaps for selecting the largest elements. For each row i this takes O(n log l_i). For the candidate list this takes O(s log k). Thus overall this takes O(∑_i n log l_i + s log k).

(iii) The best approach: Use median of medians (see: https://en.wikipedia.org/wiki/Median_of_medians) when selecting the largest elements. This avoids the logarithmic factor from sorting and guarantees a worst-case time complexity of O(mn + s). Note that QuickSelect does not work here as it is O(n^2) in the worst case. Unfortunately I think this is CS3230 material and beyond IT5003 scope.

I claim that my solution runs in $O($mn + sum(limits)$)$. See the last page for the code.

Box A.2. A Linked List Task

```
# Definition for singly-linked list.
# class ListNode:
#     def __init__(self, val=0, next=None):
#         self.val = val
#         self.next = next
class Solution:
    # you can create helper function(s)
    def reverse(node: Optional[ListNode]) -> Optional[ListNode]:
        pre = None
        cur = node
        while cur:
            nex = cur.next
            cur.next = pre
            pre = cur
            cur = nex
        return pre


    def removeNodes(self, head: Optional[ListNode]) -> Optional[ListNode]:
        # complete this
        if not head or not head.next:
            return head
        head = Solution.reverse(head)
        dum = ListNode(0, head)
        pre = dum
        cur = head
        msf = float('inf')
        while cur:
            if cur.val > msf:
                pre.next = cur.next
            else:
                msf = cur.val
                pre = cur
            cur = cur.next
        return Solution.reverse(dum.next)
```

An alternative solution via recursion: (same time complexity O(n), but space complexity is increased from O(1) to O(n) due to recursion stack)

```
def removeNodes(self, head: Optional[ListNode]) -> Optional[ListNode]:
    if not head or not head.next:
        return head
    head.next = self.removeNodes(head.next)
    if head.next.val < head.val:
        return head.next
    return head
# return the updated SLL at the end
```

I claim that my solution runs in $O($_____n_____$)$. See bottom of next page for testing.

Box A.3. Three Bags of Candies

Special case 1: return 7 * a
Special case 2: return 7 * min(a, b) + 1

Idea:
Return 7 * min((a+b+c) // 2, a+b+c - max(a, b, c)).

Justification:
Obviously maximising the score is equivalent to maximising the number of moves, M.

Notice that M is bounded by the total number of candies, T, divided by 2, as each move always removes exactly 2 candies.

Also, once all candies outside the largest bag are gone, one cannot continue. Thus M is also bounded by the sum of the number of candies of the two smaller bags, S.

Indeed, if max(a, b, c) > S, then by pairing each of the S candies with the candies from the largest bag, one gets exactly S moves. Otherwise, we are able to use up all candies if T is even, or all but one if T is odd. Thus M is exactly the minimum of S and T // 2.

Time complexity:
O(1) because each operation takes constant time, independent of a, b, c.

Code:

```
class Solution:
    def threeBagsofCandies(self, a: int, b: int, c: int) -> int:
        return 7 * min((a + b + c) // 2, a + b + c - max(a, b, c))


tests = [[1, 3, 4], [1, 2, 4], [7, 7, 1],
         [7, 5, 1], [2, 3, 5], [2, 2, 2]]
for t in tests:
    print(f"{t} -> {Solution().threeBagsofCandies(*t)}")
```

I claim that my solution runs in $O($_____1_____$)$.

Box A.4. Feedback about IT5003 S1 AY25/26 changes: Add LeetCode and Midterm

```
# P2 Testing
def build(values):
    head = None
    for v in reversed(values):
        head = ListNode(v, head)
    return head

def to_list(head):
    out = []
    while head:
        out.append(head.val)
        head = head.next
    return out
```

– END OF PAPER; All the Best –

```
tests = [[], [7], [1, 2, 3, 4], [4, 3, 2, 1], [5, 2, 13, 3, 8],
         [2, 2, 2], [10, 5, 10], [1, 1, 1, 2], [9, 1, 2, 3, 0],
         [1, 90, 19, 1, 73, 15, 70]]
for t in tests:
    print(f"{t} -> {to_list(Solution().removeNodes(build(t)))}")
```

```python
class Solution:
    from typing import List

    def gridProblem(self, grid: List[List[int]], limits:
List[int], k: int) -> int:
        def select(a, k):  # returns the kth smallest
element using median-of-medians, O(n)
            while True:
                n = len(a)
                if n <= 25:
                    return sorted(a)[k]
                m = [sorted(a[i:i + 5])[len(a[i:i + 5]) //
2] for i in range(0, n, 5)]
                p = select(m, len(m) // 2)
                l, h = [_ for _ in a if _ < p], [_ for _ in
a if _ > p]
                if k < len(l):
                    a = l
                elif k < n - len(h):
                    return p
                else:
                    k -= n - len(h)
                    a = h
        a = []
        for i, r in enumerate(grid):
            t = limits[i]
            if t == 0:
                continue
            c = select(r, len(r) - t)
            l = [x for x in r if x > c]
            a += l
            if t - len(l) > 0:
                a += [c] * (t - len(l))
        if k >= len(a):
            return sum(a)
        c = select(a, len(a) - k)
        l = [x for x in a if x > c]
        return sum(l) + c * (k - len(l))


tests = [[[[7, 7, 7], [7, 7, 7]], [1, 3], 2],
         [[[5, 10, 3, 1]], [3], 2],
         [[[1, 2, 3], [4, 5, 6], [7, 8, 9]], [3, 2, 1], 2],
         [[[22, 1, 33], [21, 4, 77]], [2, 3], 3]]
for t in tests:
    print(f"{t} -> {Solution().gridProblem(*t)}")
```